Recommended Assessment

IMU

Analyzing raw IMU data

1. When the Sensors Trainer unit is placed vertically on a surface (Z axis of the IMU pointing straight up), the Z axis measures $+9.81\ m/s^2$. Why is this value positive even though gravity is positive downwards?

2. Why would you measure and use acceleration in units of g as supposed to m/s^2?

3. Record the mean and standard deviation of the IMU data in the table below. Compare the data to noise profile for the gyroscope and accelerometer.

	Trial 1		Trial 2		Trial 3	
Sensor	μ	σ	μ	σ	μ	σ
Gyro (X)						
Gyro (Y)						
Gyro (Z)						
Accel (X)						
Accel (Y)						
Accel (Z)						

Visualizing Attitude with IMU

4. What equation did you use to find the attitude in degrees using the gyroscope data?

5. What equation did you use to find the attitude (roll & pitch) in degrees using the accelerometer data?

6. How does the roll and pitch angle estimates compare to each other from the gyroscope and accelerometer?

IMU Cheat Sheet

What is it?	
What does it measure directly?	
What can you measure indirectly?	
Where are they used?	
Pros	
Cons	
Notes/ Important things to remember	